How Plants Respond to Signals

Biology 2e chapter 9 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

A signal can change the activity of responding cells. In a plant experiment, a visible response may take days. A carrier control receives the same solvent without the signal so its effects can be distinguished from the treatment.

Investigation question

Does a growth signal change stem elongation compared with its carrier alone?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Follow three plants per treatment for seven simulated days. Measure stem height from the soil line to the shoot tip, not to the tip of a leaf.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: cm stem height

Condition / sample	Repeat	Source	cm stem height
Carrier control	1		
Carrier control	2		
Carrier control	3		
Gibberellin treatment	1		
Gibberellin treatment	2		
Gibberellin treatment	3		

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Copy this page for each trial. Treatment: _____ Repeat: ____ Source: _____

days	cm stem height	Notes
0		
1		
2		
3		
4		
5		
6		
7		

Graph your observations

Label axes and units; keep each treatment and source identifiable. For repeated measurements, connect readings only from the same sample. For ordinal scores, explain the categories instead of treating the scale as concentration.

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Compare change from each plant's starting height rather than final height alone.

Why is carrier-only a better control than an untreated plant?

Can stem height identify a receptor or prove which intracellular steps occurred?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: A whole-plant response model, not a direct receptor assay. Response depends on genotype and growing conditions. Model growth is illustrative and does not establish a classroom dose.